

6G Weekly Survey — Jul 20 - Jul 26, 2026

15 items across 4 topics, ranked by importance.

AI-native air interface

Unified Evaluation Methodology for AI-Native Integrated Sensing and Communication

Filip Lemic, Andra Blaga, Francesco Devoti, Guillermo Encinas Lago, et al. · arXiv cs.NI · 2026-07-16

★★★★■ Addresses critical gap between ISAC theory and deployment by providing standardized evaluation framework that credible research groups and 3GPP study items can adopt, though lacks direct technical innovation in PHY or algorithms.

This paper proposes a unified evaluation methodology for AI-native ISAC systems where ML agents adapt sensing, communication, and actuation policies online under uncertainty. The authors formalize the closed-loop ISAC design space and present a three-stage validation pipeline (bounds analysis, digital-twin simulation, over-the-air testing) with structured KPI-to-KVI mappings. Two representative scenarios (UAV outdoor and RIS indoor) demonstrate the framework's ability to bridge theoretical ISAC gains and deployment-ready performance claims.

Contribution: First comprehensive methodology and reporting checklist for evaluating AI-native ISAC systems that explicitly accounts for inference latency, environmental uncertainty, and the coupling between sensing accuracy, communication reliability, and resource overhead in closed-loop scenarios.

Relevance to your work: Directly relevant as it establishes evaluation standards for AI-native ISAC systems you are researching, providing methodology for comparing learned PHY approaches against classical baselines under realistic closed-loop constraints with latency and uncertainty.

LLM-Powered Agentic AI for 5G/6G Networks: A Tutorial and Survey on Architectures, Protocols, and Standardization

Mazene Ameer, Abdelkader Mekrache, Bouziane Briki, Adlen Ksentini · arXiv cs.NI · 2026-07-17

★★★★■ Solid survey that organizes emerging LLM-agent concepts for telecom, but does not present novel algorithms, data, or PHY-layer methods directly relevant to AI-native air interface design.

This tutorial-and-survey formalizes how LLM-powered agentic AI can integrate with 5G/6G control and management planes, covering reasoning, planning, tool use, and multi-agent coordination. It maps agentic capabilities onto protocol surfaces and 3GPP standardization pathways. The work bridges the gap between LLM research and telecommunications network automation, identifying open challenges for autonomous goal-driven network control.

Contribution: First systematic mapping of LLM-based agentic AI onto 5G/6G control-plane protocols and standardization bodies, moving beyond isolated treatment of AI and network domains.

Relevance to your work: Tangential—focuses on network control and management automation using LLMs rather than AI/ML at the physical layer, though the agentic coordination concepts may inform future cross-layer 6G intelligence architectures.

ISAC (sensing + communication)

Robust Design of Integrated Sensing and Communication in LEO Satellite Systems

Hezhen Yang, Xiaoming Chen, Qi Wang · arXiv cs.IT · 2026-07-14

★★★★■ Addresses critical convergence of ISAC and NTN, two priority 6G areas, with practical consideration of satellite power constraints and channel uncertainty relevant to operational systems.

This paper develops a robust beamforming algorithm for dual-function LEO satellites performing simultaneous sensing and communication. The approach minimizes transmit power under sensing MSE and communication SINR constraints while accounting for channel phase uncertainty that worsens cross-functional interference. Simulations demonstrate power savings versus baseline methods.

Contribution: First systematic framework for ISAC in LEO satellites with robust beamforming that explicitly handles channel phase uncertainty and joint resource allocation for sensing and communication tasks.

Relevance to your work: Directly combines your NTN and ISAC research interests, showing how dual-function satellite systems can manage interference and resource efficiency under realistic impairments like phase uncertainty.

Energy-Efficient Target-Aware Hybrid Beamforming for THz Near-Field ISAC with Sparse Connectivity

Nusaibah A. Alshorman, Chong Han, Huseyin Arslan · arXiv eess.SP · 2026-07-17

★★★★■ Strong technical innovation addressing a fundamental THz ISAC challenge with a counterintuitive but well-motivated approach from credible authors in sub-THz research, demonstrating practical hardware/energy benefits. This paper addresses THz ISAC's challenge of illuminating extended targets with highly directional beams by proposing a sparse-connected hybrid beamforming architecture that intentionally engineers grating lobes as controllable auxiliary beams. Instead of suppressing grating lobes, the design exploits them for single-shot full-aperture target coverage while serving multiple communication users. At 140 GHz, simulations show near fully-digital sensing accuracy with significant hardware and energy reduction compared to dense arrays.

Contribution: Novel reverse design philosophy that intentionally engineers sparse-array grating lobes as structured auxiliary illumination beams for extended-target sensing, enabling simultaneous full-aperture coverage without sequential beam scanning.

Relevance to your work: Directly relevant to your ISAC and sub-THz 6G waveform interests, offering an energy-efficient beamforming solution that combines sensing and communication with reduced hardware complexity for near-field THz scenarios.

Dual-Security for Indoor OFDM-ISAC Systems via Temporal Artificial Noise

Yinchao Yang, Yathreb Bouazizi, Prabhat Raj Gautam, Michael Breza, et al. · arXiv cs.IT · 2026-07-17

★★★★■ Addresses a critical but overlooked dual-security problem in ISAC systems with a novel temporal AN approach that exploits OFDM domain properties, highly relevant as ISAC moves toward standardization in 6G. This paper addresses dual-security in indoor OFDM-based ISAC systems where a sensing user may eavesdrop on communications and a communication user may perform unauthorized sensing. The authors propose temporal artificial noise (AN) that simultaneously degrades the eavesdropping SINR in frequency domain for malicious sensing users and increases time-domain sensing error for malicious communication users. Numerical results demonstrate effective dual protection while maintaining legitimate user performance.

Contribution: First work to simultaneously protect both communication privacy against authorized sensing users and sensing privacy against authorized communication users using a unified temporal AN scheme in OFDM-ISAC.

Relevance to your work: Directly relevant to your ISAC research interests, particularly the emerging challenge of securing both communication and sensing functions simultaneously in systems where users have dual capabilities and partial authorization.

DFT-p-FDMA Based Chirp Transmission in CP-OFDM for Unified ISAC Waveform Design

Fabrizio Carpi, Joonyoung Cho, Kyeong Jin Kim, Charlie Jianzhong Zhang · arXiv cs.IT · 2026-07-17

★★★★■ Strong contribution to ISAC waveform coexistence from a credible industry team (Qualcomm/Samsung), addressing practical integration of sensing into existing OFDM infrastructure with analytical framework and validated

performance.

The paper proposes a DFT-precoded FDMA technique to embed chirp signals within CP-OFDM systems for integrated sensing and communication, allowing radar sensing on a subset of subcarriers while maintaining interference-free OFDM data transmission on others. The method analyzes how delays and Doppler shifts affect target echoes under DFT-p-FDMA and demonstrates flexible range-Doppler resolution tuning through time-frequency resource allocation. Simulations validate delay and Doppler estimation performance in practical multi-symbol scenarios.

Contribution: First demonstration of chirp-based radar sensing embedded in CP-OFDM via DFT-precoded FDMA, enabling coexistence without mutual interference and flexible range-Doppler tradeoffs through resource allocation.

Relevance to your work: Directly relevant to your ISAC research, particularly for understanding how radar functionality can be multiplexed with communication in 5G NR/6G OFDM frameworks through waveform design and resource partitioning strategies.

Learning-Driven Channel Representation for Wireless Localization: From Channel Observations to Location Inference

Hongyu Xie, Chenglong Li, Xinming Huang, Emmeric Tanghe, et al. · arXiv eess.SP · 2026-07-16

★★★★■ Strong synthesis from credible researchers that reframes an active ISAC research area with a unifying conceptual framework, directly addressing practical challenges in generalizing learned localization models across hardware and environments.

This survey reviews learning-driven wireless localization methods through the lens of channel representation, modeling the process as a three-stage framework: wireless observation → channel representation → location inference. It systematically analyzes how different channel observation forms (CSI, CIR, AoA/ToA) are transformed into representations via feature extraction and representation learning, comparing methods by accuracy, generalization, and data requirements. The authors emphasize that channel representation quality determines localization performance and identify gaps between experimental studies and real-world deployment.

Contribution: Provides the first comprehensive survey organizing wireless localization research around channel representations as an explicit intermediate abstraction, bridging raw observations to location inference and highlighting representation quality as the decisive factor for cross-scenario generalization.

Relevance to your work: Highly relevant to your ISAC research: channel representation learning is central to both sensing (localization) and communication tasks, and the generalization challenges discussed parallel those in learned PHY designs that must adapt across deployment scenarios.

Pinching Antenna-Assisted ISAC with Waveguide Mode Selection

Ruotong Zhao, Yijia Zhang, Shaokang Hu, Derrick Wing Kwan Ng · arXiv cs.IT · 2026-07-17

★★★★■ Solid contribution advancing ISAC with a novel reconfigurable antenna concept and practical optimization framework, though limited to architectural innovation rather than fundamental physical-layer breakthroughs.

The paper introduces a mode-selectable pinching antenna (PA) framework for ISAC that jointly optimizes waveguide mode selection, transmit beamforming, and antenna positioning to maximize sensing SNR under communication QoS constraints. A block-coordinate descent algorithm with penalty-based majorization-minimization handles the mixed-integer nonconvex optimization. Simulations show substantial gains over fixed-antenna and traditional PA baselines by exploiting spatial and modal reconfigurability.

Contribution: First work to enable dynamic waveguide mode selection in PA-assisted ISAC, moving beyond static receiver assumptions to exploit additional spatial degrees of freedom for joint sensing-communication optimization.

Relevance to your work: Directly relevant to your ISAC research interests, particularly in demonstrating how reconfigurable hardware (mode-selectable PAs) can improve sensing-communication trade-offs through joint optimization of physical and signal processing parameters.

6G Demos & Trials

FLINT: Fingerprinting Federated Learning Architectures from 5G PHY-Layer Side Channels

Md Nahid Hasan Shuvo, Mahmudul Hassan Ashik, Moinul Hossain · arXiv cs.LG · 2026-07-16

★★★★■ Exposes a fundamental security gap in FL-over-5G deployments with practical over-the-air validation, directly relevant to emerging distributed AI use cases in 6G where on-device learning and privacy are critical design considerations.

This paper demonstrates a novel side-channel attack that fingerprints federated learning model architectures (CNNs, RNNs, Transformers) using only 5G PHY-layer scheduling metadata from the PDCCH, without requiring packet-level network access. The FLINT framework decodes downlink control information, tracks user devices across RNTI changes, and applies temporal pattern analysis to distinguish architecture families with 0.930 macro F1-score on an over-the-air srsRAN testbed. This reveals that encrypted 5G PHY-layer transmissions leak sufficient timing information to enable passive reconnaissance of FL clients' model types.

Contribution: First demonstration that 5G PHY-layer scheduling patterns alone (without network-layer visibility or decrypted payloads) suffice to fingerprint federated learning model architectures, revealing a new class of side-channel vulnerability in wireless AI systems.

Relevance to your work: Directly relevant to AI-native air interface design—demonstrates that PHY-layer scheduling decisions in current and future wireless standards inadvertently leak machine learning metadata, requiring careful consideration in 6G PHY design for federated or distributed AI training scenarios.

LIVE-RIS: Real-Time In-Flight Actuation of UAV-Mounted RIS

David Müller, Kevin Weinberger, Aydin Sezgin, Martin Mönnigmann · arXiv eess.SP · 2026-07-16

★★★★■ Hardware demonstration with real-time actuation addresses a critical gap between UAV-RIS theory and practice, providing essential validation data for 6G aerial infrastructure deployment.

This paper presents the first real-time UAV-mounted RIS prototype that maintains beamforming performance despite UAV motion and disturbances. The system uses the UAV's extended Kalman filter and onboard sensors to predict RIS pose and reconfigure the surface in real time. Experimental measurements validate that dynamic reconfiguration effectively compensates for disturbances and preserves link performance gains across different deployment scenarios.

Contribution: First experimental demonstration of a real-time capable UAV-mounted RIS that dynamically compensates for flight disturbances using onboard sensor fusion and EKF-based pose prediction.

Relevance to your work: Directly relevant to NTN and aerial-platform research, demonstrating practical engineering solutions for dynamic RIS control that inform design constraints for future air-ground integrated networks.

Modeling and Validation of Quality of Control for Edge-Offloaded Collaborative Navigation

Neelabhro Roy, Mikael Hammarling, Victor Nan Fernandez-Ayala, Gourav Prateek Sharma, et al. · arXiv eess.SY · 2026-07-16

★★■ This is competent systems work on 5G + robotics but remains narrowly focused on existing control theory and ROS 2 QoS policies rather than advancing wireless PHY/MAC innovation for 6G.

This paper develops a quality-of-control (QoC) framework for edge-offloaded collaborative robot navigation over 5G, modeling how wireless delay and reliability affect closed-loop control performance. The authors experimentally validate their approach on a private 5G testbed, systematically exploring interactions between control parameters (e.g., robotic motion dynamics) and network latency-reliability. Results show RELIABLE QoS provides 51.5% better QoC than BEST-EFFORT in certain scenarios, establishing control-communication co-design regimes for practical robots with dynamic collision avoidance.

Contribution: The work extends QoC theory to practical robotic systems and provides experimental validation on a real 5G testbed, quantifying the interplay between control parameters and network

performance for edge-offloaded navigation.

Relevance to your work: Tangential—addresses application-layer control over 5G networks rather than AI-native air interfaces, channel modeling, or physical-layer techniques central to beyond-5G research.

Understanding Reasoning from Pretraining to Post-Training

Jingyan Shen, Ang Li, Salman Rahman, Yifan Sun, et al. · arXiv cs.LG · 2026-07-17

★☆☆☆☆ This is pure machine learning research on language model training pipelines with no connection to wireless communications, physical layer design, or network protocols.

This paper uses chess and math tasks as controlled testbeds to study how pretraining affects reinforcement learning post-training in language models, varying model size (5M-1B parameters) and pretraining data. They find post-RL performance is predictable from pretraining loss, RL improvement rate scales linearly with pretraining tokens, and RL amplifies correct moves on easy tasks while surfacing new solutions on hard tasks. The framework provides systematic measurement of the pretraining-to-RL interface that is prohibitively expensive to study in standard LLM settings.

Contribution: First systematic, compute-controlled study quantifying how pretraining choices (data, model size) predict RL returns and revealing that RL has distinct behavioral modes (amplification vs. discovery) depending on task difficulty.

Relevance to your work: tangential

A Morphing-Designed Hexarotor Prototype combining Practical Resilience and Efficiency

Murat Bronz, Mahmoud Hamandi, Elgiz Baskaya, Chiara Gabellieri, et al. · arXiv eess.SY · 2026-07-17

★☆☆☆☆ This is UAV mechanical design research with no connection to wireless communications, physical layer techniques, channel modeling, or any 6G air interface technology.

This paper presents a hexarotor drone prototype with morphable geometry that simultaneously achieves resilience to single-propeller failures and energy efficiency comparable to standard designs. The authors experimentally validate trade-offs across continuous geometric configurations by varying inter-propeller angles, using empirical power models and failure-recovery metrics. Results demonstrate a viable geometric region where resilience does not compromise aerodynamic efficiency, with full hardware/software open-sourced.

Contribution: Experimental demonstration that morphing hexarotor geometry can achieve both practical fault resilience and energy efficiency without the typical performance trade-offs seen in redundant aerial platforms.

Relevance to your work: tangential

6G Physical Layer

Elliptic Range-Doppler Mapping for OFDM-ISAC under IQ Imbalance

Thrassos K. Oikonomou, Dimitrios Bozanis, Sotiris A. Tegos, Panagiotis D. Diamantoulakis, et al. · arXiv eess.SP · 2026-07-16

★★★★☆ Solid incremental contribution addressing a practical impairment in OFDM-ISAC with a well-motivated signal model, though scope is limited to one hardware non-ideality.

This work addresses receiver IQ imbalance in OFDM-based integrated sensing and communication systems, which creates ghost targets by coupling each subcarrier with its mirror image. Rather than first compensating IQI then detecting, the authors exploit the coupled structure directly by formulating an elliptic atom group model where each physical target induces linked direct and mirror components. They propose an elliptic group orthogonal matching pursuit detector using two weighted 2D FFTs that outperforms baselines in exact support recovery and weak-target detection under moderate to strong IQI.

Contribution: Novel direct exploitation of IQI-impaired structure through elliptic group atoms representing coupled direct/mirror components, enabling joint sparse recovery without explicit IQI compensation.

Relevance to your work: Directly relevant to your ISAC research interests, particularly for real-world deployment of OFDM-based joint sensing-communication where RF impairments like IQI degrade radar performance and must be handled at the physical layer.

Low Complexity Neural Network Digital Predistortion of Wideband Power Amplifiers through Feature Selection

Cel Thys, Rodney Martinez Alonso, Ali H Alsarraf, Dominique Schreurs, et al. · arXiv eess.SP · 2026-07-16

★★★★ Solid incremental work addressing a real deployment barrier (NN-DPD complexity) with measured validation and public dataset release, but focuses on transmitter-side linearization rather than core 6G PHY innovations.

The paper addresses computational complexity in neural network-based digital predistortion (DPD) for wideband power amplifiers by proposing a feature selection approach using LASSO and MRMR algorithms to reduce input dimensionality. Applied to measured FR3 power amplifier data (publicly released), the method achieves 30% complexity reduction compared to phase-normalized NN architectures while maintaining linearization performance comparable to polynomial baselines. The work targets practical deployment of NN-DPD by bridging the gap between research methods and implementation constraints.

Contribution: A feature-engineering pipeline combining LASSO and MRMR for systematic input pruning in NN-DPD architectures, demonstrating practical complexity reduction with measured wideband FR3 PA data released publicly.

Relevance to your work: Tangential—this is transmitter hardware optimization rather than AI-native air interface design, though general lessons on complexity-performance tradeoffs in NN-based PHY components may inform learned transmitter architectures.